



NTNU

TMA4125 Matematikk 4N

Fourier Series.

Ronny Bergmann

Department of Mathematical Sciences, NTNU.

February 3rd, 2023

Motivation for periodic functions

We would like to investigate **periodic** phenomena, for example

- ▶ alternating current
- ▶ heart beat
- ▶ water waves
- ▶ audio & audio processing

or in general phenomena that repeat.

This might also happen for complex-valued functions $f: \mathbb{R} \rightarrow \mathbb{C}$.

History – Jean-Baptiste Fourier & trigonometric series



Jean-Baptiste Joseph Fourier (1768 – 1830)

- ▶ First trigonometric series:
Euler (1750), D. Bernoulli (1753)
- ▶ Fourier (1807):
Propagation of Heat in Solid Bodies
- ▶ He postulated:

*"Every periodic function
can be written as
a superposition
of trigonometric functions"*

Periodic functions

Definition. A function $f: \mathbb{R} \rightarrow \mathbb{C}$ is called **periodic** if for some $T > 0$ it holds

$$f(x) = f(x + T) \quad \text{for all } x \in \mathbb{R}.$$

The value T is called the **period** of f .

The smallest T fulfilling the property above is called the **fundamental period** of f .

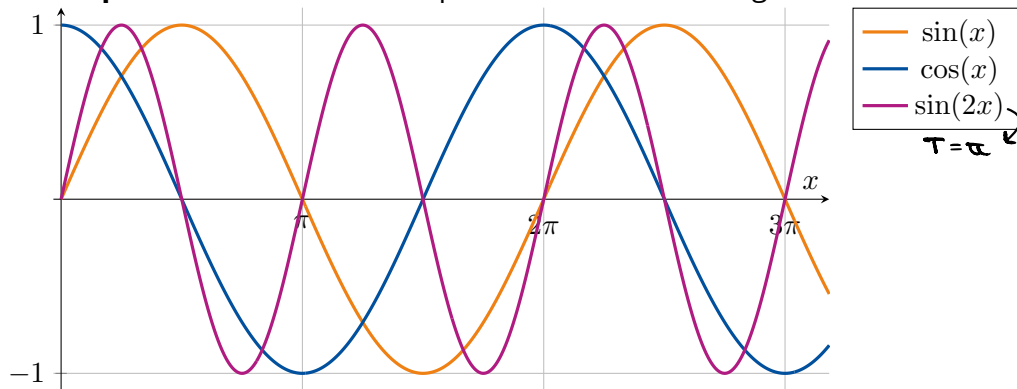
If a function g is $\frac{T}{n}$ -periodic for some $n \in \mathbb{N} \setminus \{0\}$, then the number n is also called **frequency** of g .

- ▶ Is g T -periodic? **Yes**
- ▶ What does g do in one interval of length T , e. g. $[0, T]$?

We will usually consider functions with fundamental period $T = 2\pi$.

Periodic functions – Examples.

Examples. What fundamental periods do the following functions have?

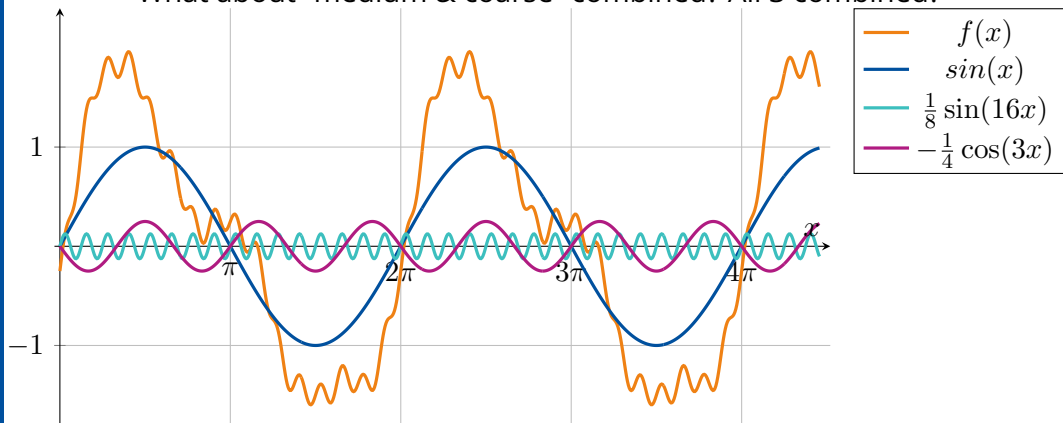


What about $f(x) = \sin(nx)$ for some $n \in \mathbb{N} \setminus \{0\}$? $T = \frac{2\pi}{n}$

Frequency in $T = 2\pi$?

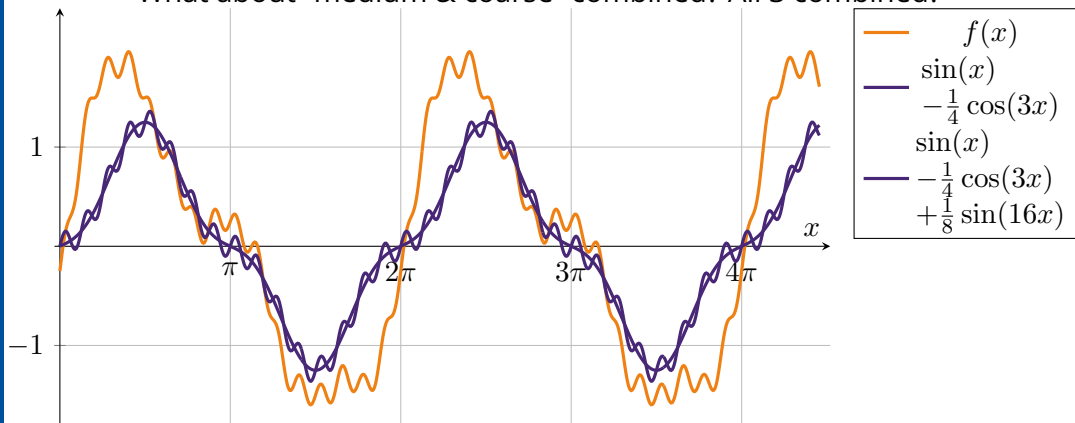
Decomposing functions

- ▶ The following function f is $T = 2\pi$ periodic and looks a little chaotic.
- ▶ Does the “coarse level look like” $\sin(x)$?
- ▶ Does the “fine level look like it wiggles” like $\sin(16x)$?
- ▶ Does the “medium scale look like” $\cos(3x)$?
- ▶ What about “medium & coarse” combined? All 3 combined?



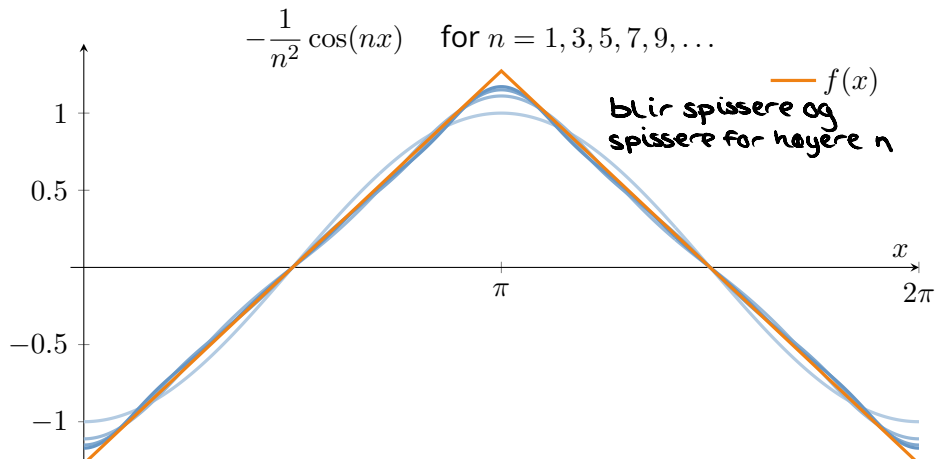
Decomposing functions

- ▶ The following function f is $T = 2\pi$ periodic and looks a little chaotic.
- ▶ Does the “coarse level look like” $\sin(x)$?
- ▶ Does the “fine level look like it wiggles” like $\sin(16x)$?
- ▶ Does the “medium scale look like” $\cos(3x)$?
- ▶ What about “medium & coarse” combined? All 3 combined?



Composing functions

Note. Given two functions f, g that are T periodic and $\alpha, \beta \in \mathbb{R}$.
Then $h(x) = \alpha f(x) + \beta g(x)$ is also T periodic.
 $\Rightarrow$ Compose functions, for example add



Fourier Series. A general construction scheme

Idea. use $\sin(nx)$ and $\cos(nx)$ for $n \in \mathbb{N}$ and $\cos(0x) = 1$

“to construct functions”.

For coefficients $a_0, a_n, b_n, n = 1, 2, \dots$ (real or complex),
we call the function

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx).$$

Forrige slide:

$$\begin{aligned} a_0 &= 0 \\ a_n &= -\frac{1}{n^2} \\ b_n &= 0 \end{aligned}$$

a **Fourier Series**.

and we denote the **Fourier partial sum** $f_N, N \in \mathbb{N}$, by

$$S_N f(x) = f_N(x) = a_0 + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx).$$

Question. What happens in the limit $\lim_{N \rightarrow \infty} f_N(x)$?

Reminder. Inner product

Let V be a complex vector space. Then a mapping $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{C}$ is called an **inner product** if the following properties hold

1. Linearity in the first argument: for all $f, g, h \in V, \alpha, \beta \in \mathbb{C}$

$$\langle \alpha f + \beta g, h \rangle = \alpha \langle f, h \rangle + \beta \langle g, h \rangle$$

2. Conjugate symmetry: for $f, g \in V$ we have

$$\langle f, g \rangle = \overline{\langle g, f \rangle}$$

if we use (2) we get:
 $\langle f, f \rangle = \overline{\langle f, f \rangle}$

3. Positive definite: for all $f \in V$ it holds

$$\langle f, f \rangle \geq 0 \quad \text{and} \quad \langle f, f \rangle = 0 \Leftrightarrow f = 0$$

Example. Let V be the space of complex continuous 2π periodic functions. $f: [0, 2\pi) \rightarrow \mathbb{C}$. Then

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) \overline{g(x)} \, dx$$

is a complex inner product on V .

Reminder. Euler formula

- ▶ Euler's formula states $e^{ix} = \cos(x) + i \sin(x)$
- ▶ plugging in $-x$ yields $e^{-ix} = \cos(x) - i \sin(x)$
- ▶ Adding both yields

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

- ▶ Subtracting both yields

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

- ▶ and we also have e. g.

$$\cos(nx) = \frac{e^{inx} + e^{-inx}}{2} = \frac{(e^{ix})^n + (e^{-ix})^n}{2}$$

⇒ these are hence also called **trigonometric polynomials**

- ▶ **But keep in mind the representation as e^{ix} is a complex function.**

The space of trigonometric polynomials up to degree N .

We denote by

then $g: [-\alpha, \alpha] \rightarrow \mathbb{R}$

later $\mathbb{R}$

$$V_N := \left\{ g \mid g = a_0 + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx), \quad a_0, a_1, \dots, a_N, b_1, \dots, b_N \in \mathbb{C} \right\}$$

the (real) vector space of periodic functions of degree at most N (or functions that can be “composed” by sine and cosine up to frequency N)

With the equations from last slide we can also consider

$$\tilde{V}_N := \left\{ g \mid g = \sum_{k=-N}^N c_k e^{ikx}, \quad c_{-N}, c_{-N+1}, \dots, c_N \in \mathbb{C} \right\}$$

the complex vector space of trigonometric polynomials of degree at most N .

...and see directly that $V_N = \tilde{V}_N$.

A norm on a complex vector space

Observation. As for a real vector space, the complex inner product introduces a **norm** on V given by

$$\|f\| = \sqrt{\langle f, f \rangle}$$

Remember that we have the **Cauchy-Schwarz inequality**

$$|\langle f, g \rangle| \leq \|f\| \|g\|$$

Question. When/how can we “describe” a 2π -periodic function f “best possible” by a function $g \in V_N$, i. e. such that

$$\|g - f\| \quad \text{is as small as possible (among all } g \in V_N\text{)?}$$

Orthogonal and orthonormal systems

A sequence/family $\{\varphi_n\}_n$ of functions is called

► orthogonal if

$$\langle \varphi_k, \varphi_n \rangle = \begin{cases} 0 & \text{if } k \neq n \\ d_n \neq 0 & \text{if } k = n, \end{cases}$$

► orthonormal if

$$\langle \varphi_k, \varphi_n \rangle = \begin{cases} 0 & \text{if } k \neq n \\ 1 & \text{if } k = n, \end{cases}$$

Note. All functions φ_k are normed since $\|\varphi_k\| = \sqrt{\langle \varphi_k, \varphi_k \rangle} = 1$

⇒ We can create an orthonormal system from an orthogonal one by rescaling $\tilde{\varphi}_k = \frac{1}{\sqrt{d_k}} \varphi_k$

The trigonometric system is orthonormal

Remember. for (complex) 2π periodic functions: $\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) \overline{g(x)} \, dx$.

Example 1. the sequence $\varphi_k = e^{ikx}$, $k \in \mathbb{Z}$, is an orthogonal system.

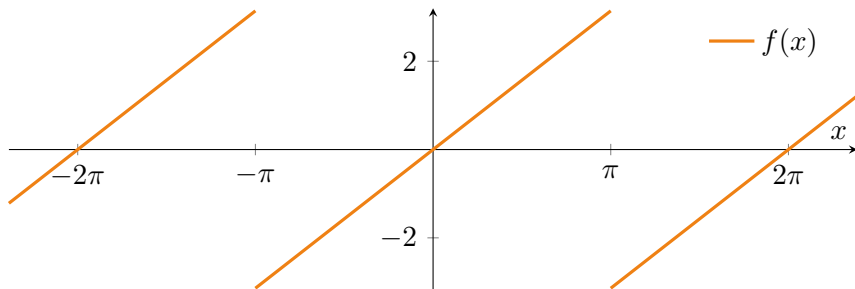
Example 2. We consider the family $\{1, \cos(x), \sin(x), \cos(2x), \sin(2x), \dots\}$ or shorter $\{\sin(kx)\}_{k=1}^{\infty} \cup \{\cos(kx)\}_{k=0}^{\infty}$.
These functions also form a orthogonal system

How to “best describe” f using V_N ?

Let f be a 2π -periodic function, which function is “closest” to f from within V_N , i. e. which function $g \in V_N$ minimises $\|g - f\|$?

Example. We can define a 2π periodic function f by just defining it on one interval of length 2π and continue the rest:

$f(x) = x$, for $x \in [-\pi, \pi)$ and periodically continued.



...and how to best compute g (i. e. coefficients a_n, b_n or the c_k s)?

The simpler case: $g = f \in V_N$ – Compute the coefficients.

Assume. In the following let V be a complex vector space with inner product $\langle \cdot, \cdot \rangle$ and an orthogonal system $\{\varphi_k\}_k \subset V$.

$\Rightarrow V_N := \left\{ g = \sum_{k=1}^N d_k \varphi_k \right\}$ is a N -dimensional subspace spanned by the first N members of $\{\varphi_k\}_k$.

Proposition. A function $f \in V_N$ is of the form

$$f(x) = \sum_{k=1}^N c_k \varphi_k.$$

And we can compute the coefficients by

$$c_k = \frac{\langle f, \varphi_k \rangle}{\|\varphi_k\|^2}.$$

Even shorter. If $\{\varphi_k\}_k$ is an **orthonormal** system, then $c_k = \langle f, \varphi_k \rangle$.

(General) Fourier coefficients

Definition. For an orthogonal system $\{\varphi_k\}_k \subset V$ and a given vector/function $f \in V$ the coefficients

$$c_k = \frac{\langle f, \varphi_k \rangle}{\|\varphi_k\|^2}.$$

are called (general) **Fourier coefficients**.

Sometimes we write $\hat{f}(k) = c_k$

Orthogonal Projection & best approximation

For $f \in V$ we define the **projection** $\Pi_N f \in V_N$ by

$$\Pi_N f = \sum_{k=1}^N c_k \varphi_k \quad \text{with} \quad c_k = \frac{\langle f, \varphi_k \rangle}{\|\varphi_k\|^2}.$$

Observations.

- ▶ Π_N is a linear mapping from $f \in V$ to $\Pi_N f \in V_N$
- ▶ Π_N is indeed a projection, that is $\Pi_N f = f$ if f is already in V_N
- ▶ even more: $\Pi_N f$ is orthogonal in the sense that **for any** $g \in V_N$ we have

$$\langle f - \Pi_N f, g \rangle = 0$$

Theorem. Let $f \in V$ be given. Then $\Pi_N f \in V_N$ satisfies

$$\|f - \Pi_N f\| = \min_{g \in V_N} \|f - g\|.$$

General Fourier Series

Definition. (General Fourier series for an orthogonal system)

Let $\{\varphi_k\}_{k=1}^{\infty}$ be an orthogonal system. Then the formal expression

$$\sum_{k=1}^{\infty} \hat{f}(k) \varphi_k, \quad \hat{f}(k) = \frac{\langle f, \varphi_k \rangle}{\|\varphi_k\|^2}$$

is called a **general Fourier series**.

Since $\Pi_N f = \sum_{k=1}^N \hat{f}(k) \varphi_k$ is the partial sum, so we write $S_N f = \Pi_N f$.

For the specific case for periodic functions we still need

$$V := \left\{ f: [-\pi, \pi) \rightarrow \mathbb{C} \mid \int_{-\pi}^{\pi} |f(x)|^2 dx < \infty \right\},$$

where we assume that the integral exists.

Functions $f \in V$ are said to be **square-integrable**.

V is indeed a vector space.

Complex trigonometric series / Fourier series

Definition. (Complex Fourier series)

Let f be a 2π -periodic function and consider the orthogonal system

$$\{e^{ikx}\}_{k \in \mathbb{Z}}$$

on $[-\pi, \pi)$.

Then the formal series

$$\sum_{k=-\infty}^{\infty} c_k e^{ikx}, \quad \text{with } c_k = \frac{\langle f, e^{ikx} \rangle}{\|e^{ikx}\|^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

is called the **complex Fourier series** associated with f .

We also write $c_k(f)$ or $\hat{f}(k)$ for the c_k .

We write this association also as

$$f \sim \sum_{k=-\infty}^{\infty} c_k e^{ikx}$$

take $k \in \mathbb{N}$, then we
had the best approx thm

Real trigonometric series / Fourier series

Definition. (Real Fourier series)

Let f be a 2π -periodic function and consider the orthogonal system

$$\{1\} \cup \{\cos(nx)\}_{n \in \mathbb{N}} \cup \{\sin(nx)\}_{n \in \mathbb{N}}$$

$= \cos(0)$

"Compute the
real Fourier series"

Then the formal series associated to f given by

$$f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

Underscore writing

with coefficients

$$a_0 = \frac{\langle f, 1 \rangle}{\langle 1, 1 \rangle} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx,$$

$$a_n = \frac{\langle f, \cos(nx) \rangle}{\|\cos(nx)\|^2} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{\langle f, \sin(nx) \rangle}{\|\sin(nx)\|^2} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

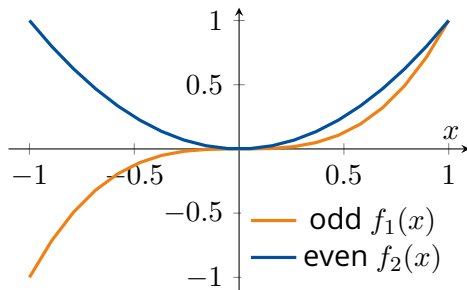
is called the **(real) Fourier series** associated with f .

Note. a_0 and $a_n, b_n, n \in \mathbb{N}$, are all real values $\Leftrightarrow f$ is real valued.

Reminder. Even and odd functions

A function is called **even** if $f(x) = f(-x)$ holds for all x .

A function is called **odd** if $f(x) = -f(-x)$ holds for all x .



we have the following

f	g	$f \cdot g$
even	even	even
odd	even	odd
even	odd	odd
odd	odd	even

Examples. For $n \in \mathbb{N}$: $\cos(nx)$, is even, $\sin(nx)$ is odd.

Even better.

$\sin^2(nx) \Rightarrow \text{even}$

▶ if f is odd then $\int_{-L}^L f(x) dx = 0$

▶ if f is even then $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$

Relations between the real and complex Fourier series

We can use Euler's formula to relate the (complex) Fourier coefficients c_k and the (real) Fourier coefficients a_0, a_n, b_n :

Proof below

$$a_0 = c_0$$

$$a_n = c_n + c_{-n}, \quad n = 1, \dots,$$

$$b_n = i(c_n - c_{-n}), \quad n = 1, 2, \dots,$$

$$c_n = \frac{a_n - ib_n}{2}, \quad n = 1, \dots,$$

$$c_{-n} = \frac{a_n + ib_n}{2}, \quad n = 1, 2, \dots,$$

$$\left. \begin{array}{l} a_n = c_n + c_{-n} \\ b_n = i(c_n - c_{-n}) \end{array} \right\} a_n + ib_n$$

and even more: the projection onto $V_N = \tilde{V}_N$ yield the same partial sum

$$S_N(f) = \sum_{k=-N}^N c_k e^{ikx} = a_0 + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx).$$

felles envelope å rekne ut

⇒ Choose the coefficients that "fit better".

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{i0x} dx = c_0$$

$$\begin{aligned} c_n + c_{n-1} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) (e^{inx} + e^{-inx}) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \left(\frac{e^{inx} + e^{-inx}}{2} \right) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = a_n \end{aligned}$$

Ek 0: $f(x) = 4 \cos(6x)$

$$\therefore a_0 + \sum_{n=1}^{100} a_n \cos(nx) + \underbrace{b_n}_{=0} \sin(nx)$$

ingen konstant ledd i $f(x)$

$a_6 = 4$

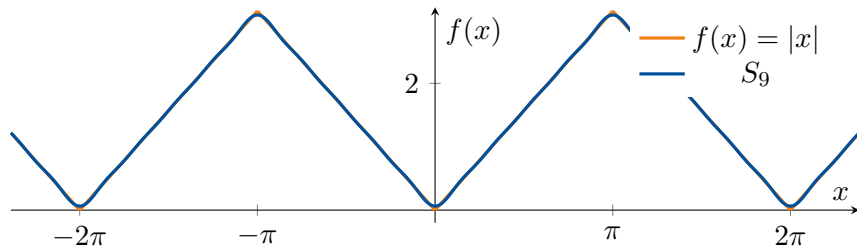
ingen sin-ledd i $f(x)$

By comparing coefficients we get

$$a_0 = 0 \quad a_n = \begin{cases} 0 & n \neq 6 \\ 4 & n = 6 \end{cases} \quad b_n = 0 \quad \text{for all } n$$

Samme prinsipp for $g(x) = 17 \sin(13x)$ og $h(x) = 7e^{-8ix}$

Example 1. Fourier coefficients of $f(x) = |x|$, $x \in [-\pi, \pi)$.



Question. Does $S_N f$ tend to f for $N \rightarrow \infty$? And if so, in what sense?

It will get very close, not always the case

$$f(x) = |x|$$

$f(x)$ is even

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)| dx = \frac{2}{2\pi} \int_0^{\pi} x dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \overset{\text{even} \cdot \text{even} = \text{even}}{f(x) \cos(nx) dx} = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx = \left[-\frac{x \sin(nx)}{n} - \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi n^2} (\cos(n\pi) - 1) = \frac{2}{\pi n^2} ((-1)^n - 1)$$

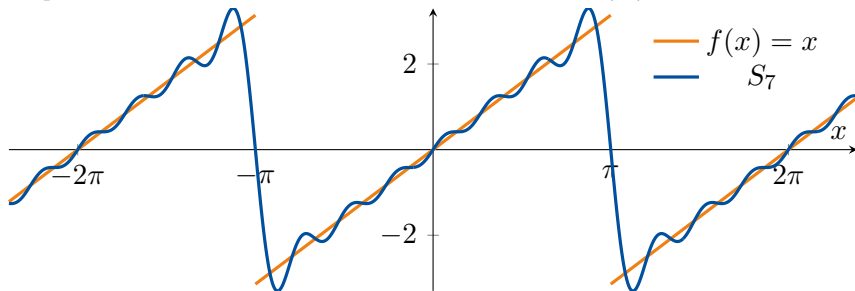
$\begin{cases} \frac{-4}{\pi n^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \overset{\text{even} \cdot \text{odd} = \text{odd}}{f(x) \sin(nx) dx} = \dots = 0$$

$$\text{We computed } f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} ((-1)^n - 1) \cos(nx)$$

Complex Fourier coefficients of $f(x) = x$



Here we can ask the same question about convergence as well.

odd function $\Rightarrow a_n = 0$ & $a_0 = 0$ (krystall-origo)

$$f \sim \sum_{k=-\infty}^{\infty} c_k e^{ikx}$$

For $k \neq 0$:

$$c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-ikx} dx$$

derivative $\leadsto$

$$= \frac{1}{2\pi} \left(-\frac{1}{ik} x e^{-ikx} \Big|_{-\pi}^{\pi} + \frac{1}{ik} \int_{-\pi}^{\pi} e^{-ikx} dx \right)$$

$$= \frac{1}{2\pi} \left(-\frac{1}{ik} x e^{-ikx} \Big|_{-\pi}^{\pi} + \frac{1}{(ik)^2} e^{-ikx} \Big|_{-\pi}^{\pi} \right)$$

$$= \frac{e^{ik\pi} + e^{-ik\pi}}{2ik} - \frac{1}{k^2} \cdot 2i \underbrace{\left(\frac{e^{-ik\pi} + e^{ik\pi}}{2i} \right)}_{\substack{\sin(-\pi k) = 0 \\ 0}}$$

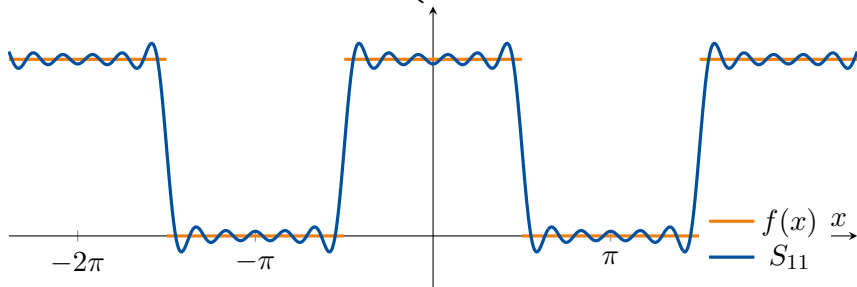
$$= \frac{1}{ik} \cos(k\pi)$$

$$= \begin{cases} -\frac{1}{ik} & k \text{ even} \\ \frac{1}{ik} & k \text{ odd} \end{cases} = \frac{(-1)^{k-1}}{ik}$$

$$c_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0$$

(Shifted) Heaviside function

Consider the function $f(x) = \begin{cases} 1 & \text{for } |x| \leq \frac{\pi}{2} \\ 0 & \text{else.} \end{cases}$



...but let's turn to Python to really see the effect.

even function $\Rightarrow b_n = 0$

$$a_0 = \frac{1}{2}$$

Convergence – Motivation

We introduced the Fourier series as

$$f \sim \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ikx}$$

with

$$\hat{f}(k) = \langle f, e^{ikx} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx$$

Question. When/For which x does

$$f(x) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ikx} = \lim_{N \rightarrow \infty} S_N f(x)$$

hold?

Remember that $S_N f$ is the **Fourier partial sum**

$$S_N f(x) = \sum_{k=-N}^N \hat{f}(k) e^{ikx}$$

The Dirichlet kernel

The N th Dirichlet kernel is defined by

$$D_N(x) = \sum_{k=-N}^N e^{ikx} = 1 + 2 \sum_{n=1}^N \cos(nx).$$

$a_0 = 1$
 $a_n = \begin{cases} 2 & \text{for } n \leq N \\ 0 & \text{else} \end{cases}$

$c_k(D_N) = \begin{cases} 1 & \text{for } |k| \leq N \\ 0 & \text{else} \end{cases}$

And since $\int_{-\pi}^{\pi} \cos(nx) dx = 0$ for any $n = 1, 2, \dots$ we get

$$\int_{-\pi}^{\pi} D_N(x) dx = 2\pi.$$

We can define the **normalized Dirichlet Kernel**

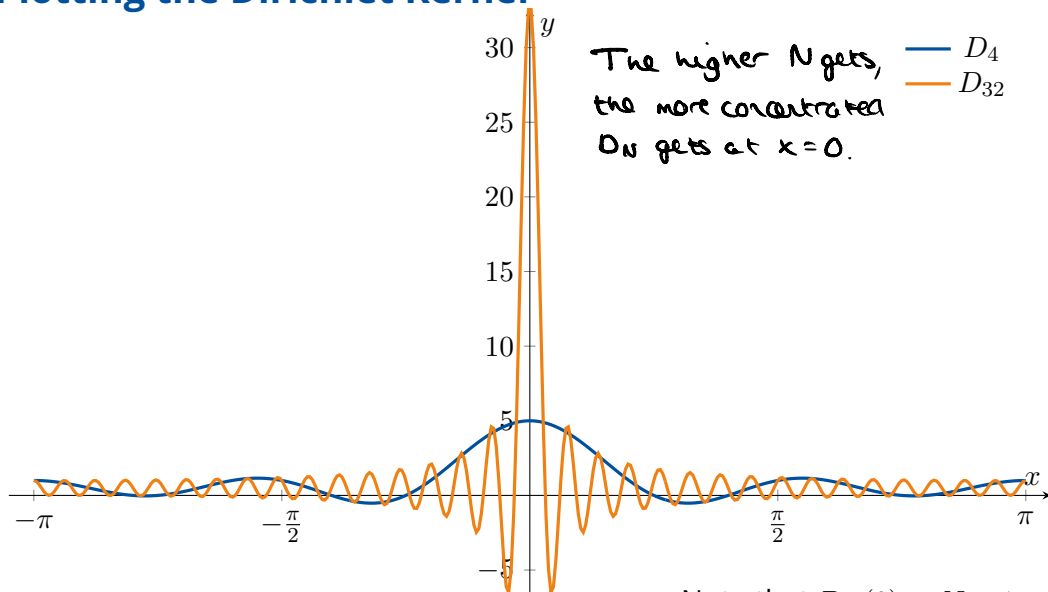
$$\tilde{D}_N(x) = \frac{1}{2\pi} D_N(x) \quad \text{where we get} \quad \int_{-\pi}^{\pi} \tilde{D}_N(x) dx = 1$$

Rewriting the Dirichlet kernel

Using a shift in the sum index and the geometric sum with $q = e^{ix}$ we can rewrite

$$D_N(x) = \sum_{k=-N}^N e^{ikx} = \frac{\sin((N + \frac{1}{2})x)}{\sin(\frac{x}{2})}$$

Plotting the Dirichlet Kernel



Note that $D_N(0) = N + 1$

Convolution

Definition. For two 2π -periodic functions f, g the **convolution** (norsk.: **konvolusjon**, dt.: Faltung) is denoted by $f * g$ and defined by

$$(f * g)(x) = \int_{-\pi}^{\pi} f(y)g(x - y) dy.$$

Lemma. For two 2π -periodic functions it holds that

$$(f * g)(x) = \int_{-\pi}^{\pi} f(y)g(x - y) dy = \int_{-\pi}^{\pi} g(y)f(x - y) dy = (g * f)(x).$$

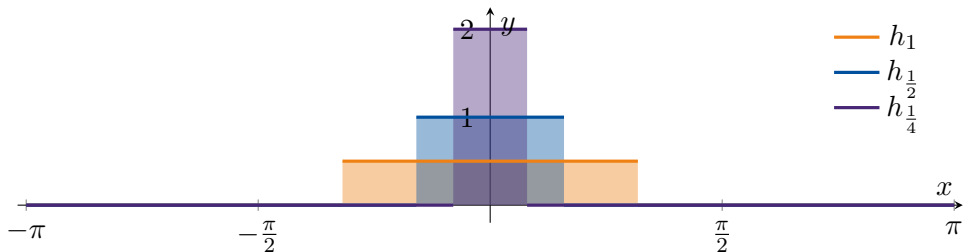
Proof. $(f * g)(x) = \int_{-\pi}^{\pi} f(y)g(x - y) dy$ substitute $t = x - y$, $trg! = x - y \rightarrow g(t) = x - t$
 $= - \int_{x+\pi}^{x-\pi} f(x - t)g(t) dt$ $\frac{dt}{dy} = -1 \Rightarrow dt = -dy$
 $= \int_{-\pi}^{\pi} f(x - t)g(t) dt$ $\rightarrow \int_{x+\pi}^{x-\pi} = \int_{x-\pi}^{x+\pi}$
 $= (g * f)(x)$ $\rightarrow$ function is 2π -periodic

Understanding Convolution – Averaging

Let's convolve our function f first with a special function to understand convolutions better.

For $0 < \varepsilon < \pi$ let h_ε be the 2π periodic function given by

$$h_\varepsilon(x) = \frac{1}{2\varepsilon} X_{[-\varepsilon, \varepsilon]}(x) = \begin{cases} \frac{1}{2\varepsilon} & \text{for } |x| \leq \varepsilon \\ 0 & \text{else.} \end{cases}$$



Note that we have (similar to the dirac idea): $\int_{-\pi}^{\pi} h_\varepsilon(x) dx = 1.$

Convolving with h_ε (at $x = 0$)

When we now look at $f * h_\varepsilon$ we get

$$(f * h_\varepsilon)(0) = \frac{1}{2\varepsilon} \int_{-\varepsilon}^{\varepsilon} f(y) dy$$

This is an average value of f on ε -“neighborhood” around 0.

If f is “nice enough” we would expect $(f * h_\varepsilon)(0) \rightarrow f(0)$ as $\varepsilon \rightarrow 0$.

If f is continuous this follows directly from the fundamental theorem of calculus:

$$\begin{aligned} (f * h_\varepsilon)(x) &= \int_{-\varepsilon}^{\varepsilon} f(y) h_\varepsilon(x-y) dy \\ &= \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(y) dy && \text{definition} \\ &= \frac{1}{2\varepsilon} (F(x+\varepsilon) - F(x-\varepsilon)) && \varepsilon \rightarrow 0 \approx f(x) \\ &\quad \frac{1}{2} \end{aligned}$$

Convolution and Fourier Coefficients

Lemma. Let f, g be 2π -periodic functions, that are absolutely integrable. Then we can relate convolution and Fourier coefficients as

$$c_k(f * g) = 2\pi c_k(f)c_k(g)$$

Proof. (see notes)

Note. Some books define the convolution with a factor $\frac{1}{2\pi}$ upfront, then it vanishes on the right hand side.

Note 2. with this Lemma, we can write $S_N f = \tilde{D}_N * f$

Pointwise Convergence of Fourier Series

Theorem.

Let f be a 2π periodic function that is piecewise continuously differentiable such that at a jump the left and right hand side limits

$$f(x^-) := \lim_{y \rightarrow x^-} f(y) \quad \text{and} \quad f(x^+) := \lim_{y \rightarrow x^+} f(y)$$

as well as the left and right derivatives at the jumps x

$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

exist. Then the Fourier series converges to the midpoint

$$\lim_{N \rightarrow \infty} S_N f(x) = \frac{1}{2} (f(x^-) + f(x^+))$$

At points where f is continuous, we obtain **pointwise convergence** to f

$$\lim_{N \rightarrow \infty} S_N f(x) = \frac{1}{2} (f(x^-) + f(x^+)) = f(x) \text{ on } x \in (-\pi, \pi)$$

Proof. We omit the proof here.

Fourier Series on $[-L, L]$ – Idea

We can also consider functions of period $2L$, $L > 0$.

Idea. “Stretch/Squeeze” all formulae we had until now.

Example. since $\sin(x)$ is 2π -periodic, $\sin(\frac{\pi}{L}x)$ is $2L$ -periodic.

Then

$$\left\{ e^{\frac{ikx\pi}{L}} \right\}_{k \in \mathbb{Z}} \quad \text{and} \quad \left\{ \cos \frac{n\pi x}{L} \right\}_{n=0}^{\infty} \cup \left\{ \sin \frac{n\pi x}{L} \right\}_{n=1}^{\infty}$$

are **orthogonal systems** on $[-L, L]$ with respect to the scalar product

$$\langle f, g \rangle = \int_{-L}^L f(x) \overline{g(x)} \, dx$$

Fourier Series on $[-L, L]$

The Fourier series read

$$f \sim \sum_{k=-\infty}^{\infty} c_k e^{\frac{ikx\pi}{L}}, \quad c_k = \frac{1}{2L} \int_{-L}^L f(x) e^{-\frac{ikx\pi}{L}} dx$$

and

$$f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}$$

with coefficients

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx, \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx.$$

Even and odd extensions

Definition. (nearly a reminder)

A function f on a symmetric interval $I = [-L, L]$, $L > 0$, is called

- ▶ is called **even** if $f(x) = f(-x)$ holds for all $x \in I$.
- ▶ is called **odd** if $f(x) = -f(-x)$ holds for all $x \in I$.

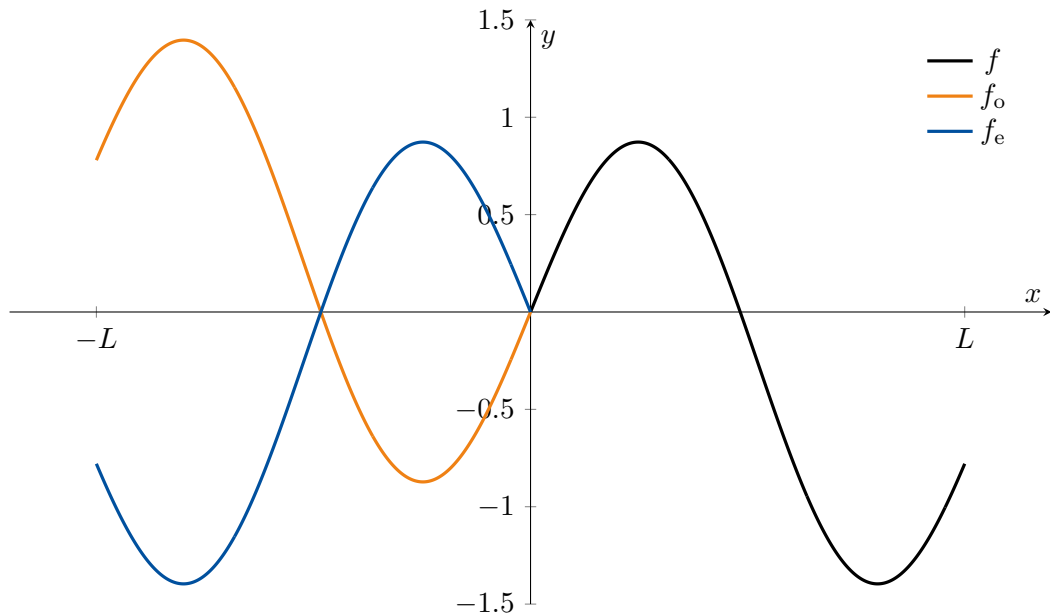
These mean that it is enough to know a function on **half** the interval and we can extend it

Definition. For $f: [0, L] \rightarrow \mathbb{R}$ we define its

- ▶ **odd extension** $f_o(x) = \begin{cases} f(x) & \text{if } x \in [0, L) \\ -f(-x) & \text{if } x \in (-L, 0) \end{cases}$
- ▶ **even extension** $f_e(x) = \begin{cases} f(x) & \text{if } x \in [0, L) \\ f(-x) & \text{if } x \in (-L, 0) \end{cases}$

Note. Both f_o and f_e extend f from $[0, L]$ to $[-L, L]$.

Illustration of even and odd extensions



Observation for extensions

Observe that we can compute the Fourier series of f_e and f_o by just using f .

For f_o we get

$$a_0(f_o) = 0 \quad a_n(f_o) = 0 \quad b_n(f_o) = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

or in other words a sine series

For f_e we get

$$a_0(f_e) = \frac{2}{L} \int_0^L f(x) dx \quad a_n(f_e) = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx \quad b_n(f_e) = 0$$

or in other words a cosine series

Parseval Identity

Theorem.

Let f be given such that $\int_{-L}^L |f(x)|^2 dx$ exists and is finite (or similarly: $f \in L^2((-L, L))$). Let

$$\begin{aligned} f &\sim \sum_{k=-\infty}^{\infty} c_k e^{\frac{ikx\pi}{L}} \\ &\sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \end{aligned}$$

Then

$$\frac{1}{2L} \|f\|^2 = \frac{1}{2L} \int_{-L}^L |f(x)|^2 dx = \sum_{k=-\infty}^{\infty} |c_k|^2 = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Proof.

Application of Parsevals Identity

For the heaviside function $u(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$ defined on $x \in [-\pi, \pi]$ we can compute its Fourier series

$$u(x) \sim \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin((2n-1)x)$$

and u is smooth in $x = \frac{\pi}{2}$ and satisfies the assumptions of the theorem for pointwise convergence.

Thus

$$1 = u\left(\frac{\pi}{2}\right) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin\left((2n-1)\frac{\pi}{2}\right)$$

Note that the sine alternates between ± 1 at the points. Rearranging yields

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{2} \left(1 - \frac{1}{2}\right) = \frac{\pi}{4}$$

Spectrum of periodic functions

For a $2L$ -periodic function we defined its Fourier series as

$$f \sim \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ikx \frac{\pi}{L}}$$

where

$$\hat{f}(k) = c_k = \frac{1}{2L} \int_{-L}^L f(x) e^{-ikx \frac{\pi}{L}} dx \in \mathbb{C}.$$

We can associate with f a sequence of pairs $(c_k, \frac{k}{2L})_{k \in \mathbb{Z}}$ which is called the **spectrum of f** .

Amplitude

Since $\hat{f}(k) = c_k \in \mathbb{C}$ we can also write this complex number as

$$c_k = |c_k|e^{i\theta_k},$$

where $|c_k| \in \mathbb{R}$ is the **amplitude** and θ_k is the **phase**.

The number $\frac{1}{2L}$ is called the frequency.

Further properties

If we have two 2π -periodic functions f, g with associated Fourier series

$$f \sim \sum_{k \in \mathbb{Z}} c_k(f) e^{ikx} \quad \text{and} \quad g \sim \sum_{k \in \mathbb{Z}} c_k(g) e^{ikx}.$$

The the following properties hold

- ▶ **real functions and $c_k(f)$** if f is real-valued, then $c_k(f) = \overline{c_{-k}(f)}$
- ▶ **Linearity.** the Fourier coefficients of $h_1(x) = \alpha f(x) + \beta g(x)$ ($\alpha, \beta \in \mathbb{R}$) are given by $c_k(h_1) = \alpha c_k(f) + \beta c_k(g)$ for $k \in \mathbb{Z}$.
- ▶ **Translation** of a function: For $h_2(x) = f(x - x_0)$ for some $x_0 \in [-\pi, \pi)$ yields the Fourier coefficients are $c_k(h_2) = e^{ikx_0} c_k(f)$.
- ▶ **Modulation** of a function: For $h_3(x) = e^{-ik_0 x} f(x)$ for some $k_0 \in \mathbb{Z}$ the Fourier coefficients are $c_k(h_3) = c_{k+k_0}(f)$.
- ▶ **Differentiation.** if f is absolutely continuous and both f, f' are in L_1 (their absolute value ins integrable) then

$$c_k(f') = 2\pi i k c_k(f)$$